ARENLAR

Aromatik uglevodorodlar · umumiy formula C_nH_{2n-6}

Student book – nazariy qo‘llanma

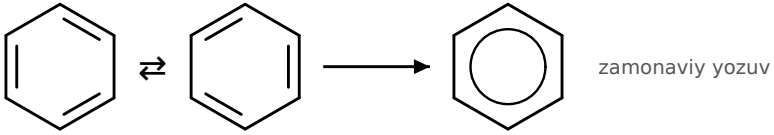
Molekulasida **benzol halqasi** bo‘lgan karbotsiklik birikmalar **aromatik uglevodorodlar** yoki **arenlar** deyiladi. Qatorning birinchi vakili – **benzol** $\mathbf{C_6H_6}$. Benzol halqasi soniga qarab arenlar **bir halqali** (benzol, toluol, ksilollar) va **ko‘p halqali** (naftalin $C_{10}H_8$, antratsen $C_{14}H_{10}$) turlarga bo‘linadi. Bir halqali arenlarning umumiy formulasi $\mathbf{C_nH_{2n-6}}$ ($n \geq 6$): ular mos alkandagidan 8 ta vodorod atomi kam.

1. Tuzilishi

Benzol formulasining tarixi

1865-yilda **A. Kekule** benzolni uchta qo‘sh bog‘i bor, bir-birini almashtirib turuvchi olti a’zoli halqa deb ifodaladi. Bu formula benzolning kimyoviy xossalarini to‘liq tushuntirib bera olmadi:

- benzol qo‘sh bog‘ saqlagan birikmalarga xos **bromli suvni** va $\mathbf{KMnO_4}$ eritmasini rangsizlantirmaydi;
- birikish reaksiyalari qiyin, **o‘rin olish** reaksiyalari esa oson boradi;
- benzolning dialmashingan hosilasi (masalan, dibrombenzol) Kekule bo‘yicha 4 ta emas, faqat **3 ta** izomerga ega;
- halqadagi barcha C–C bog‘lar **bir xil** uzunlikda (0,140 nm), oddiy va qo‘sh bog‘lar almashinmaydi.

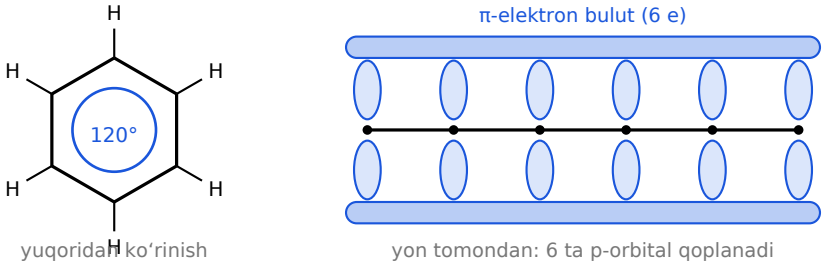


Kekule formulalari → aromatik halqa (aylana – delokalizatsiyalangan π-elektronlar)

Zamonaviy tasavvur

- Benzolning 6 ta uglerod atomi ham **sp^2-gibridlangan**. Barcha σ-bog‘lar bir tekislikda yotadi, valent burchak **120°**, molekula **yassi** (tekis olti burchak).
- Har bir uglerodda halqa tekisligiga perpendikulyar **bittadan p-orbital** qoladi. 6 ta p-orbital yon tomonlari bilan qoplanib, halqaning tepasi va tagida yagona **π-elektron bulut** hosil qiladi.

- π-elektronlar (6 ta) 6 ta uglerod atomiga **delokalizatsiyalangan** (tekis taqsimlangan), shuning uchun benzolda oddiy va qo‘sh bog‘lar yo‘q: har bir C–C bog‘ oraliq (bir yarim) xarakterga ega.
- Delokalizatsiya molekulani juda **barqaror** qiladi. Bu xossa **aromatiklik** deyiladi.



Benzol molekulasining orbital tuzilishi

Xyukkel qoidasi: birikma aromatik bo‘lishi uchun molekula **tekis halqali**, π-elektronlar **delokalizatsiyalangan** va ularning soni **4n + 2** (n = 0, 1, 2, ...) bo‘lishi kerak. Benzol: n = 1, π-elektronlar 6 ta. Naftalin: n = 2, π-elektronlar 10 ta.

Bog‘larning taqqoslanishi

Modda	Bog‘	Uzunligi, nm
etan	C–C	0,154
benzol	C–C (aromatik)	0,140
etilen	C=C	0,134
atsetilen	C≡C	0,120

Bog‘ uzunligi ortib borishi: C≡C < C=C < C–C (benzol) < C–C (etan).

Umumiy formulalar (C_nH_{2n-6})

Ko‘rsatkich	Ifoda	Misol: toluol C_7H_8
Molyar massa	$M_r = 14n - 6$	92
Vodorod bo‘yicha zichlik	$D(H_2) = 7n - 3$	46
Vodorod atomlari soni	$2n - 6$	8
Jami atomlar soni	$3n - 6$	15
σ-bog‘lar soni	$3n - 6$	15
Delokalizatsiyalangan π-bog‘lar	3 (6 ta elektron)	3
sp^2 / sp^3 uglerod atomlari	$6 / n - 6$	6 / 1
Yonishi	$C_nH_{2n-6} + (1{,}5n - 1{,}5)O_2 \rightarrow nCO_2 + (n - 3)H_2O$	$C_7H_8 + 9O_2 \rightarrow 7CO_2 + 4H_2O$

Gomologik qator

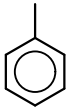
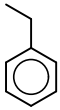
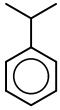
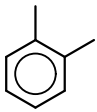
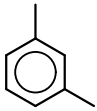
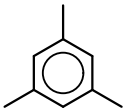
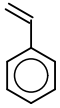
Formula	Nomi	Formula	Nomi
C_6H_6	benzol	C_9H_{12}	propilbenzol, kumol
C_7H_8	toluol	$C_{10}H_{14}$	butilbenzollar
C_8H_{10}	etilbenzol, ksilollar	C_nH_{2n-6}	...

Stirol $C_6H_5–CH=CH_2$ (C_8H_8) benzol gomologi emas: yon zanjirida qo‘sh bog‘ bor, formulasi C_nH_{2n-8}.

2. Nomlanishi

Arenlar uch xil nomlanadi: **tarixiy**, **ratsional** (orto-, meta-, para-) va **sistematik** (IUPAC).

A) Tarixiy (trivial) nomlar

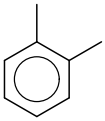
Formula	Skelet	Sistematik	Tarixiy
C_6H_6		benzol	benzol
C_6H_5–CH_3		metilbenzol	toluol
C_6H_5–C_2H_5		etilbenzol	etilbenzol
C_6H_5–$CH(CH_3)_2$		izopropilbenzol	kumol
$C_6H_4(CH_3)_2$ (1,2-)		1,2-dimetilbenzol	o-ksilol
$C_6H_4(CH_3)_2$ (1,3-)		1,3-dimetilbenzol	m-ksilol
$C_6H_4(CH_3)_2$ (1,4-)		1,4-dimetilbenzol	p-ksilol
$C_6H_3(CH_3)_3$ (1,3,5-)		1,3,5-trimetilbenzol	mezitilen
C_6H_5–$CH=CH_2$		vinilbenzol	stirol

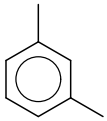
B) Ratsional nomenklatura – o-, m-, p- qo‘shimchalari

Dialmashingan benzol hosilalarida o‘rinbosarlarning o‘zaro joylashishi old qo‘shimcha bilan ko‘rsatiladi:

- **orto-** (o-): o‘rinbosarlar yonma-yon uglerodlarda (1,2-holat);
- **meta-** (m-): bitta uglerod atomi orqali ajralgan (1,3-holat);

- **para-** (p-): ikkita uglerod atomi orqali ajralgan, bir-biriga qarama-qarshi (1,4-holat).

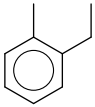

o-ksilol (1,2-)

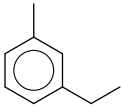

m-ksilol (1,3-)

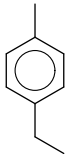

p-ksilol (1,4-)

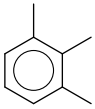
C) Sistematik nomenklatura (IUPAC)

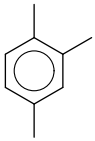
- Asos qilib **benzol** olinadi, o‘rinbosarlar (radikallar) benzol so‘zi oldidan yoziladi.
- Benzol halqasi uglerodlari o‘rinbosarlarga **eng kichik raqamlar** tushadigan qilib raqamlanadi.
- Raqamlar teng bo‘lsa, **alifbo bo‘yicha birinchi** radikal 1-raqam oladi. Radikallar alifbo tartibida aytiladi, di-, tri- qo‘shimchalari hisobga olinmaydi.
- Bitta o‘rinbosarli hosilada raqam yozilmaydi (metilbenzol).
- Yon zanjir uzun yoki qo‘sh bog‘li bo‘lsa, halqa **fenil** (C_6H_5–) radikali sifatida nomlanadi.

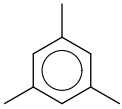

1-etil-2-metilbenzol


1-etil-3-metilbenzol


1-etil-4-metilbenzol


1,2,3-trimetilbenzol


1,2,4-trimetilbenzol


1,3,5-trimetilbenzol

D) Muhim radikallar

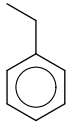
Radikal	Formula	Qaysi birikmadan
fenil	C_6H_5–	benzol
benzil	C_6H_5–CH_2–	toluol (yon zanjirdan)
tolil (o-, m-, p-)	CH_3–C_6H_4–	toluol (halqadan)
aril (Ar–)	benzol halqali radikallarning umumiy nomi	arenlar

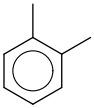
3. Izomeriyasi

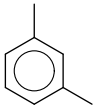
Arenlarda **yon zanjir** izomeriyasi, o‘rinbosarlarning **halqadagi holati** bo‘yicha va **sinflararo** izomeriya uchraydi. Fazoviy (sis-trans) izomeriya bo‘lmaydi.

1) Yon zanjir va o‘rinbosar holati izomeriyasi

C_8H_{10} tarkibli arenlar – **4 ta** izomer (etilbenzol va 3 ta ksilol):

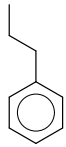

etilbenzol

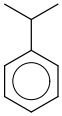

o-ksilol

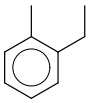

m-ksilol

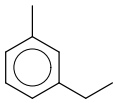

p-ksilol

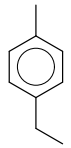
C_9H_{12} tarkibli arenlar – **8 ta** izomer:

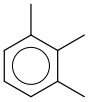

propilbenzol

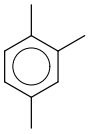

izopropilbenzol (kumol)

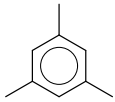

o-metiletilbenzol


m-metiletilbenzol


p-metiletilbenzol


1,2,3-trimetilbenzol


1,2,4-trimetilbenzol


1,3,5-trimetilbenzol

2) Sinflararo izomeriya

Arenlar C_nH_{2n-6} umumiy formulali **alkinlar**, **alkadiyenlar** va poliyenlar bilan izomer. Masalan, C_6H_6 tarkibli: benzol, geksadiin-1,5 ($HC{\equiv}C–CH_2–CH_2–C{\equiv}CH$), geksadiin-2,4 ($CH_3–C{\equiv}C–C{\equiv}C–CH_3$).

4. Fizik xossalari

- Arenlar – **suyuq** (oddiy sharoitda ba’zilari qattiq: naftalin) o‘ziga xos o‘tkir hidli moddalar.
- Suvda **erimaydi**, suvdan yengil (benzol zichligi 0,88 g/sm^3), organik erituvchilarda yaxshi eriydi, o‘zi ham yaxshi erituvchi.
- Molekulyar massa ortishi bilan **qaynash harorati ortadi**.
- Izomerlarda: radikallar bir-biriga yaqin bo‘lsa, qaynash harorati yuqoriroq (o- > m- > p-). **Simmetrik** tuzilgan izomerning suyuqlanish harorati yuqori (p-ksilol qotadi).

Modda	Formula	Agregat holati (20 °C)	t_{suyuq}, °C	$t_{qaynash}$, °C
benzol	C_6H_6	suyuq	5,5	80,1
toluol	$C_6H_5CH_3$	suyuq	–95	110,6
etilbenzol	$C_6H_5C_2H_5$	suyuq	–95	136
o-ksilol	$C_6H_4(CH_3)_2$	suyuq	–25	144
m-ksilol	$C_6H_4(CH_3)_2$	suyuq	–48	139
p-ksilol	$C_6H_4(CH_3)_2$	suyuq	13	138
stirol	$C_6H_5CH{=}CH_2$	suyuq	–31	145
naftalin	$C_{10}H_8$	qattiq	80	218

Ehtiyot bo‘ling: benzol va uning bug‘lari **zaharli** (qonga ta’sir qiladi, saraton kasalligini keltirib chiqarishi mumkin). Arenlar bilan faqat shamollatiladigan joyda ishlanadi.

5. Olinishi

Arenlar sanoatda **koks gazi** (toshko‘mir smolasi) va **neft** mahsulotlaridan ajratib olinadi, shuningdek quyidagi usullarda sintez qilinadi.

1) Sikloalkanlarni degidrogenlash (N.D. Zelinskiy)

